# Mohammed Eid Abdelmeguid

AI Engineer | Machine Learning Engineer

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## Professional Summary

AI/ML engineer with experience in Python backend development, retrieval-augmented generation (RAG), large language model (LLM) applications, and AI agent workflows. Contributed proxy-mode and document-ingestion features during an AI internship at Sprints. Built multi-agent applications and Arabic NLP systems; co-authored an ArabicNLP 2026 paper reporting blind-test results for spoken visual question answering.

## Technical Skills

**Programming and Data:** Python, SQL, Pandas, NumPy, data cleaning, preprocessing, PostgreSQL
**Generative AI and Retrieval:** RAG, LangGraph, AI agents, document ingestion, chunking, embeddings, semantic search, pgvector, Qdrant, structured LLM outputs
**Machine Learning and Evaluation:** Scikit-learn, XGBoost, TensorFlow/Keras, feature engineering, PCA, grid search, classification, retrieval evaluation (Recall@K, NDCG@K, MAP); familiarity with PyTorch
**NLP and Multimodal AI:** Arabic NLP, CAMeL Tools, named entity recognition, BERTopic, knowledge graphs (NetworkX), Whisper speech recognition, vision-language models
**Backend Engineering:** FastAPI, Django, Flask, REST APIs, Celery, Redis, SQLAlchemy/SQLModel, unit testing
**Deployment and Tools:** Docker, AWS EC2, Vercel, Git, GitHub Actions, Linux, Ollama, Groq API

## Professional Experience

**AI Intern | Sprints**
Oct 4, 2026 - Nov 15, 2026
*SprintFlow Digital Twin - Proxy Mode and RAG Data Ingestion*

- Implemented document ingestion from file upload through parsing, chunking, embedding, and indexing for PDF, DOCX, plain-text, and Markdown files.
- Built access-aware document retrieval with source citations using FastAPI, LangGraph, PostgreSQL, and Qdrant.
- Developed owner-controlled proxy replies and the knowledge used to ground them, with conversation-thread continuity and scoped access to private context.

**Full Stack Developer Intern | NfrtiX**
Mar 2025 - Apr 2025

- Built an internal Django and JavaScript application to support operational workflows.

**Freelance Full Stack Developer | StockGo, Qatar**
Nov 2024 - Apr 2025

- Built and deployed the complete StockGo website, including frontend and backend, for a live marketplace connecting buyers and sellers of surplus construction materials and equipment.
- Developed Django REST APIs and a PostgreSQL-backed analytics dashboard. Live website: thestockgo.com/en/.

**Backend Developer Intern | Venture Appital**
Oct 2024 - Feb 2025

- Developed modular Flask REST microservices and unit tests.

## Projects

**AI Compass | Multi-Agent RAG and Recommendation Platform**
May 2026 - Present

- Led a team of five building an AI news platform with seven ingestion sources, five enrichment agents, semantic search, and a streaming RAG assistant.
- Implemented pgvector candidate retrieval, weighted ranking, and MMR diversification with an offline NDCG@10/MAP evaluation harness; integrated Django, Celery/Redis, and Groq-served LLMs.
- Deployed the frontend on Vercel and backend on AWS EC2 with PostgreSQL.

**AynVQA-MSA | Arabic Spoken Visual Question Answering**
Jul 2026 - Oct 2026

- Implemented a modular Whisper speech-recognition, schema-constrained parsing, conditional repair, and joint vision-language answer-selection pipeline.
- Co-authored published results: **65.6% accuracy on 1,000 blind-test examples**, versus 79.0% on 500 development examples; analyzed the recording-condition shift and a caching reproducibility issue.

**Palestinian Cultural Knowledge Platform | Arabic NLP and Retrieval**
Jun 2026 - Aug 2026

- Owned ingestion, Arabic NLP, retrieval, and knowledge-graph work on a two-person project; built quality scoring, normalization, and persistent MinHash/LSH deduplication.
- Published 882 Arabic/English documents (about 890K words) on Hugging Face with gated file access; built a graph of 13,063 canonicalized entities and a retriever reporting Recall@5 of 0.93 over 1,282 passages.

**Fitness Activity Tracker | Sensor-Based Machine Learning**
Mar 2026 - Apr 2026

- Preprocessed wearable sensor data, engineered PCA/FFT features, compared five classifiers across five feature sets, and implemented independent repetition counting on a three-person team.

## Publications

**Digilians at ImageEval 2026 Shared Tasks: Modular Speech Parsing and Joint Multimodal Reasoning for Arabic Spoken VQA**
Ahmed Eid Hassan, Nour Eldeen Hossam, Ahmed Hossam Rashad, Abdelrhman Mahmoud Fawzy, **Mohammed Eid Abdelmeguid**, and Mohamed Hamdy Mohamed. *ArabicNLP 2026: Shared Tasks*, pp. 766–770, Oct 2026. DOI: 10.18653/2026.arabicnlp-sharedtasks.96.

## Education

**AI and Data Science Diploma | Digilians**
MCIT / Egyptian Military Academy scholarship | Dec 2025 - Sep 2026

**BS in Communication and Electronics Engineering | Alexandria University**
Faculty of Engineering | Sep 2019 - Jun 2024 | GPA: 3.13/4.0
ADAS graduation project: A+, ranked 5th of about 60 in the Valeo competition. Led software architecture and computer vision; mentored by Valeo and sponsored by ITIDA's Egypt Makes Electronics program.

## Certifications

**AWS Certified Machine Learning Engineer - Associate** | Amazon Web Services | Sep 2026
**Huawei Certified ICT Associate - AI (HCIA-AI)** | Huawei | Aug 2026
**IBM AI Engineering Professional Certificate** | IBM / Coursera | Jun 2026
**Machine Learning Specialization** | Stanford / DeepLearning.AI | May 2026
**Generative AI for Data Science with Copilot** | Microsoft / Coursera | Apr 2026
**Microsoft Data Engineer (159 hours)** | DEPI | Oct 2024